

Project Plan

WBS, Gantt chart against the concession deadline and resource histogram

Deliverable 6 of 6 · Case Study no. 39 · PortFlow

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Case	PortFlow - Container Terminal Yard and Gate Operations		

1. Schedule Basis

The plan uses the expected durations from Report 1: critical path A-B-D-F-G = **31 weeks**, deadline **34 weeks**, giving a 3-week buffer. The case study gives activities only as letters; the work names below are an **assumed mapping** and can be renamed without changing the numbers.

Act	Work package	Weeks (TE)	Predecessor	Team size
A	Requirements and architecture	6	-	4
B	Planning engine build (yard allocation, crane sequencer): highest uncertainty	7	A	5
C	Slot appointment module	4	A	3
D	Execution layer and equipment integration	8	B	5
E	Gate and portal integration	5	C	3
F	Integration, simulation and replay testing	7	D, E	6
G	Pilot, training and go-live	3	F	4

Team sizes are planning assumptions. Red rows are on the critical path.

2. Work Breakdown Structure (WBS)

The WBS breaks the project into eight work packages (1.1-1.8) that map to the activities above. It is shown in two parts for readability.

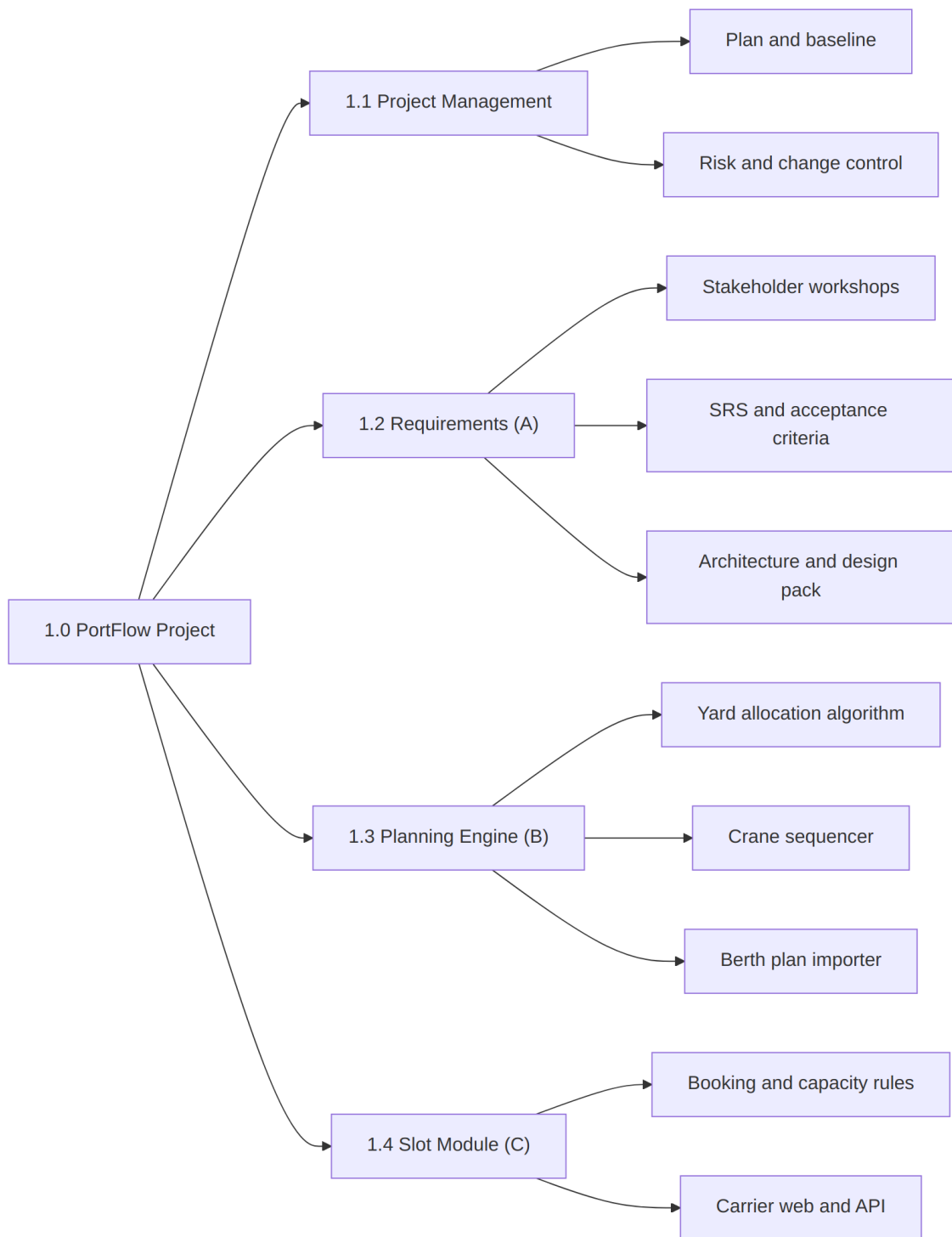


Figure 1a: WBS, part 1 (work packages 1.1-1.4)

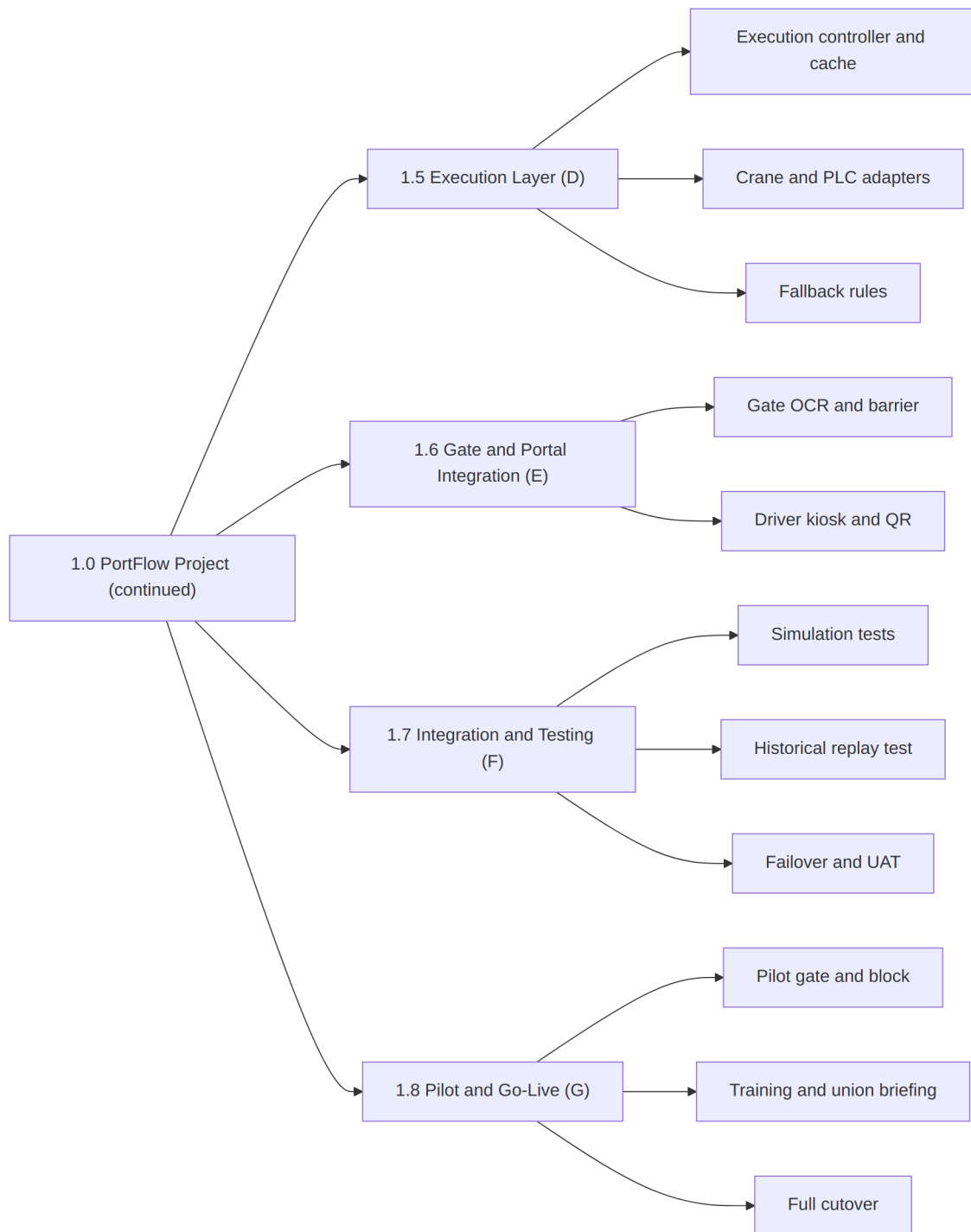


Figure 1b: WBS, part 2 (work packages 1.5-1.8)

3. Gantt Chart against the Concession Deadline

Assumed start: Monday 2 November 2026. Expected finish = week 31 (7 June 2027). Concession deadline = week 34 (28 June 2027).

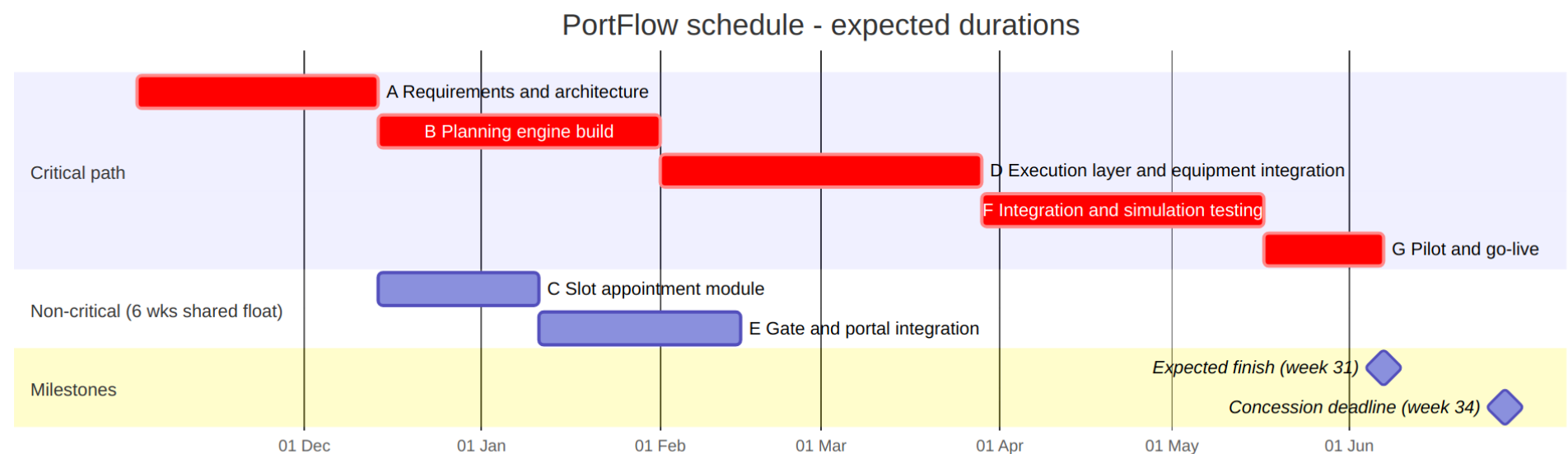


Figure 2: Gantt chart. Critical path in red; C and E carry 6 weeks of shared float; two milestones mark expected finish and deadline

4. Milestones

Milestone	Week	Date (assumed start 2 Nov 2026)
Requirements and architecture baselined (A done)	6	14 Dec 2026
Planning engine complete (B done)	13	1 Feb 2027
Execution layer and integration complete (D done)	21	29 Mar 2027
Integration and replay testing complete (F done)	28	17 May 2027
Go-live (G done), expected finish	31	7 Jun 2027
Concession deadline	34	28 Jun 2027

5. Resource Histogram

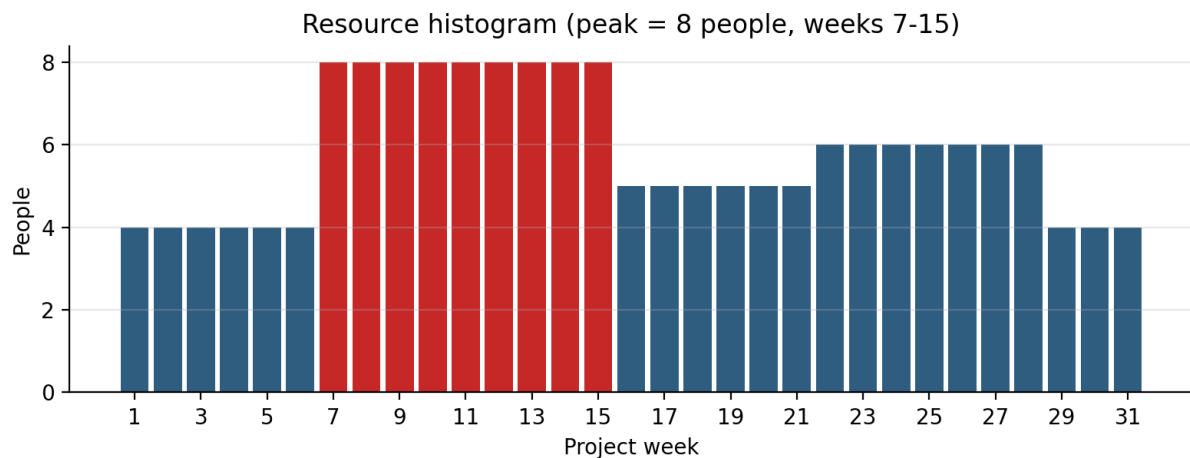


Figure 3: People required per week. Peak of 8 people in weeks 7-15

Weeks	Active tasks	People
1-6	A	4
7-10	B + C	8
11-13	B + E	8
14-15	D + E	8
16-21	D	5
22-28	F	6
29-31	G	4

Resource levelling: C and E have float, so they can be shifted within their shared 6-week window to flatten the peak of 8 without delaying the project.

6. Monitoring and Control

- Weekly critical-path review. Activities B, D and F are watched most closely: zero float and the highest variance (B and F hold 6.78 of the 7.78 total variance).
- The forecast finish is recalculated weekly. **If the forecast exceeds 34 weeks, the crash trigger applies: crash B (up to 3 weeks, ₹21 L), then F (up to 2 weeks, ₹18 L); never D.**
- Monthly risk review (see Risk Register); change control board for any scope change.
- Progress reported against baseline using earned schedule on critical-path activities.